Abdallah Elbohy

Full Stack Developer (React & FastAPI) with AI Engineering Background

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SUMMARY

Full stack developer with an MSc in Computer Engineering (BAU, Istanbul) who builds responsive React front ends and Python REST APIs with FastAPI, backed by SQL databases and vector search. Freelance data scientist since 2018 with hands-on AI engineering experience (LLM, RAG and multi-agent systems), so features ship end to end: from the database, through the API, to the screen. Comfortable with Git, Docker, CI/CD with GitHub Actions, and Azure and Google Cloud.

SKILLS

Front end: React, React Router, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, CSS Modules, Responsive / Mobile-First Design, REST API integration, Vite

Back end: Python, FastAPI, REST API design, Streamlit, Docker, Kubernetes, Airflow

Databases & Search: SQL / MySQL, Supabase, FAISS, Qdrant, Azure AI Search

AI & ML: LLM-RAG systems, LangChain, LangGraph, NLP, OCR, PyTorch, TensorFlow, Scikit-learn, Pandas, Hugging Face

Cloud & DevOps: Azure, Google Cloud, Terraform, GitHub Actions (CI/CD), GitHub Pages, Git/GitHub, npm, Langfuse

EXPERIENCE

Junior Data Scientist (Freelance) — Upwork, Remote 2018 – Present

Designing and delivering machine learning and data-driven solutions for international clients, turning business problems into working models.

- Built and deployed ML/DL models for diverse client projects.
- Managed end-to-end data workflows: collection, cleaning, scaling and modeling.
- Delivered key projects including breast cancer clustering and market text classification.
- Earned a consistent 5/5 client rating by meeting deadlines and exceeding expectations.

NLP Engineering Intern — Hattya, Bahrain 2019

Supported the development of NLP solutions and OCR pipelines for Arabic text processing.

- Built and optimized OCR systems to digitize and analyze text documents.
- Cleaned, preprocessed and annotated large-scale text datasets.
- Trained and validated NLP models and helped integrate them into live workflows.

FULL STACK & AI PROJECTS

Multi-Agent BelgeNavi TR 2024 – 2025

Python, FastAPI, Streamlit, LangChain/LangGraph, FAISS/Qdrant, Docker · github.com/Abdallaheng2022/BelgeNavi-Ikemat-assistant

- AI assistant guiding expatriates in Türkiye through residence permit renewal, address change and company formation, in Turkish, English and Arabic.
- FastAPI back end with a web UI; multi-step agent that classifies the question, retrieves, cites sources and builds a checklist.
- Checklists list required documents, fees and official links, each step tied to its source.
- Integrated Bright Data Web Search for search and scraping on Google, Bing and Reddit; prompted an SLM (Qwen3-4B) to extract facts from official documents.

TrialMatch — Clinical Trial Matcher 2025

Python, FastAPI, Streamlit, FAISS · github.com/Abdallaheng2022/Trial-and-matcher-assitant

- Ranks ClinicalTrials.gov studies for a short patient profile as Likely, Unlikely or Need more info, with short reasons and official trial links.
- REST endpoint (POST /match) built with FastAPI and a Streamlit front end.
- Pipeline: intake, vector search, eligibility parsing, rule-based matching and explanation, with guardrails on every answer.

Multi-modal Medical Assistant — Document QA App 2024 – 2025

Python, Streamlit, LangChain, FAISS, Azure AI Search, Azure Document Intelligence, OpenAI · github.com/Abdallaheng2022/Multi-modal_Medical_Assistant

- Streamlit app that answers questions about uploaded medical PDFs, scans and images such as X-rays.
- OCR-to-LLM pipeline: Azure Document Intelligence extraction, GPT-4o clean-up and image understanding.
- Hybrid chunking (paragraphs and tables plus fixed windows) to solve context fragmentation.
- Switchable vector store, FAISS locally or Azure AI Search in the cloud, with top-k tuning.

FRONT-END PROJECTS

Developer Portfolio — React & TypeScript 2026

React, TypeScript, Tailwind CSS, Vite, GitHub Actions · abdallaheng2022.github.io/abdallah-portfolio-Full-stack

- Responsive portfolio with project filters, saved light/dark theme and keyboard-friendly navigation.
- CI/CD with GitHub Actions: every push to main is type-checked, built and deployed to GitHub Pages.

All Around the World — Countries Explorer 2026

React 19, React Router, Tailwind CSS, REST API · github.com/Abdallaheng2022/All_Around_WORLD

- Single-page app listing every country with live data from a public REST API, cached in the browser for faster repeat visits.
- Search by name, filter by region, and dynamic routes that open country details without a page reload.
- Light/dark theme switch with the choice saved.

Calorie Tracker — React App 2026

React 19, React Router, CSS Modules, localStorage · github.com/Abdallaheng2022/Calorie_Tracker

- Meal logging with a modal form and live validation driven by useReducer.
- Shared state through React Context, nested routes, and records saved in localStorage.

EDUCATION

M.Sc. Computer Engineering — Bahcesehir University (BAU), Istanbul, Türkiye 2026

Thesis: Multilingual LLM mind-map generation from text (English, Arabic, Turkish): up to 91.9% quality, beat PaLM-2 coverage baseline, 35–39% faster comprehension.

B.Sc. Computer Science — Helwan University, Cairo, Egypt 2018

Thesis: Arabic & English Poetry Meter Detection using RNNs (with Cornell collaboration).

Courses

Three practical Gen AI courses by Dr. Ahmed Sallab: Agents, RAG, Prompt Engineering, Multi-modality conversions.